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Proposed Directional Forecasting Update for the Padma Bridge ML Project

Mawa- and Jajira-Side Traffic and Toll Revenue Forecasting

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1 Introduction

The existing Padma Bridge machine-learning system predicts the total daily traffic and total toll revenue of the bridge. However, the total values do not explain how traffic is distributed between the Mawa and Jajira sides.

Directional forecasting will extend the current system by separately predicting the traffic volume and toll revenue recorded at both sides of the bridge. This is particularly useful during Eid and major public holidays because the dominant direction of travel may change before and after an event.

The proposed system will produce separate Mawa- and Jajira-side forecasts while maintaining consistency with the existing total traffic and toll forecasts.

2 Objectives

The objectives of the proposed directional forecasting update are:

- To predict traffic recorded at the Mawa side of the Padma Bridge.
- To predict traffic recorded at the Jajira side of the Padma Bridge.
- To predict toll revenue separately for the Mawa and Jajira sides.
- To study how the directional distribution of traffic changes before, during and after Eid.
- To compare different machine-learning models for directional forecasting.
- To ensure that the sum of the directional forecasts remains consistent with the predicted bridge-wide total.
- To support directional forecasts for the next day, next seven days and next thirty days.

3 Current System

The current system predicts two bridge-wide quantities:

$$\hat{Y}_{t+1}^{\text{traffic}} = \text{Predicted total traffic}, \quad (1)$$

$$\hat{Y}_{t+1}^{\text{toll}} = \text{Predicted total toll revenue}. \quad (2)$$

Although the input dataset contains `Traffic_Mawa`, `Traffic_Jajira`, `Cash_Mawa` and `Cash_Jajira`, these variables are not currently used as separate prediction targets.

Therefore, the existing total forecasting models can be preserved, but new directional models must be trained.

4 Directional Forecasting Targets

The updated system will predict the following four targets:

$$\hat{Y}_{t+h}^{M,T} = \text{Predicted Mawa-side traffic,} \quad (3)$$

$$\hat{Y}_{t+h}^{J,T} = \text{Predicted Jajira-side traffic,} \quad (4)$$

$$\hat{Y}_{t+h}^{M,C} = \text{Predicted Mawa-side toll revenue,} \quad (5)$$

$$\hat{Y}_{t+h}^{J,C} = \text{Predicted Jajira-side toll revenue,} \quad (6)$$

where h is the forecasting horizon. For the proposed multi-horizon system:

$$h = 1, 2, \dots, 30. \quad (7)$$

The main outputs are summarized below.

Table 1: Directional forecasting targets

Target	Unit	Meaning
Mawa-side traffic	Vehicles	Number of vehicles recorded at the Mawa side.
Jajira-side traffic	Vehicles	Number of vehicles recorded at the Jajira side.
Mawa-side toll	BDT	Toll revenue collected at the Mawa side.
Jajira-side toll	BDT	Toll revenue collected at the Jajira side.

The interpretation of Mawa and Jajira as specific travel directions should be verified from the original Bangladesh Bridge Authority data definitions before the outputs are labelled as northbound or southbound.

5 Need for New Model Training

The existing models were trained using total traffic and total toll revenue as their targets. They cannot automatically predict the individual Mawa and Jajira values.

The existing relationship is:

$$X_t \longrightarrow \left(\hat{Y}_{t+1}^{\text{total traffic}}, \hat{Y}_{t+1}^{\text{total toll}} \right). \quad (8)$$

The directional system must learn:

$$X_t, h \longrightarrow \left(\hat{Y}_{t+h}^{M,T}, \hat{Y}_{t+h}^{J,T}, \hat{Y}_{t+h}^{M,C}, \hat{Y}_{t+h}^{J,C} \right). \quad (9)$$

Therefore, new directional models must be trained. The existing total traffic and toll models should not be deleted because their predictions can be used to improve the consistency of the directional outputs.

6 Proposed Forecasting Strategies

Two directional forecasting strategies should be compared.

6.1 Strategy 1: Direct Directional Forecasting

Separate regression models will directly predict each target:

$$\hat{Y}_{t+h}^{M,T} = f_{M,T}(X_t, h), \quad (10)$$

$$\hat{Y}_{t+h}^{J,T} = f_{J,T}(X_t, h), \quad (11)$$

$$\hat{Y}_{t+h}^{M,C} = f_{M,C}(X_t, h), \quad (12)$$

$$\hat{Y}_{t+h}^{J,C} = f_{J,C}(X_t, h). \quad (13)$$

This strategy is simple and allows each side to learn its own historical and event-related pattern.

However, independently predicted side values may not exactly equal the predicted bridge-wide total.

6.2 Strategy 2: Total-and-Share Forecasting

The recommended approach is to preserve the existing total forecasts and train additional models to predict the proportion assigned to each side.

The historical Mawa traffic share is defined as:

$$S_t^{M,T} = \frac{Y_t^{M,T}}{Y_t^{M,T} + Y_t^{J,T}}. \quad (14)$$

The Jajira traffic share is:

$$S_t^{J,T} = 1 - S_t^{M,T}. \quad (15)$$

After predicting the total traffic and Mawa share, the directional traffic forecasts are calculated as:

$$\hat{Y}_{t+h}^{M,T} = \hat{Y}_{t+h}^{\text{total traffic}} \times \hat{S}_{t+h}^{M,T}, \quad (16)$$

$$\hat{Y}_{t+h}^{J,T} = \hat{Y}_{t+h}^{\text{total traffic}} \times (1 - \hat{S}_{t+h}^{M,T}). \quad (17)$$

A similar method can be used for toll revenue:

$$\hat{Y}_{t+h}^{M,C} = \hat{Y}_{t+h}^{\text{total toll}} \times \hat{S}_{t+h}^{M,C}, \quad (18)$$

$$\hat{Y}_{t+h}^{J,C} = \hat{Y}_{t+h}^{\text{total toll}} \times (1 - \hat{S}_{t+h}^{M,C}). \quad (19)$$

This method guarantees:

$$\hat{Y}_{t+h}^{M,T} + \hat{Y}_{t+h}^{J,T} = \hat{Y}_{t+h}^{\text{total traffic}}, \quad (20)$$

$$\hat{Y}_{t+h}^{M,C} + \hat{Y}_{t+h}^{J,C} = \hat{Y}_{t+h}^{\text{total toll}}. \quad (21)$$

Therefore, the total-and-share method is recommended as the primary directional forecasting strategy.

7 Input Features

The directional forecasting models will use the existing project features together with new side-specific features.

7.1 Existing Inputs

- Previous total Padma Bridge traffic;
- Previous total toll revenue;
- Total traffic and toll lags;
- Rolling traffic and toll statistics;
- Previous Jamuna Bridge traffic and toll information;
- Day of the week and month;
- Weekend and public-holiday indicators;
- Eid indicator;
- Number of days before or after Eid;
- Temperature, rainfall, humidity and wind speed;
- Railway-operation indicator;
- Forecast horizon.

7.2 New Directional Inputs

The following side-specific features should be added:

- Mawa traffic lags of 1, 2, 3, 7, 14 and 28 days;
- Jajira traffic lags of 1, 2, 3, 7, 14 and 28 days;
- Mawa and Jajira traffic rolling averages;
- Mawa and Jajira traffic rolling standard deviations;
- Mawa and Jajira toll-revenue lags;
- Historical Mawa traffic share;
- Historical Jajira traffic share;
- Historical Mawa toll share;
- Historical Jajira toll share;
- Lagged directional difference;
- Lagged revenue per vehicle for each side.

The directional traffic difference may be defined as:

$$D_t = Y_t^{M,T} - Y_t^{J,T}. \quad (22)$$

A positive value indicates that Mawa-side traffic is higher, while a negative value indicates that Jajira-side traffic is higher.

All traffic, toll and share features must be shifted by at least one day before they are used as model inputs.

8 Models to Be Compared

The following models should be trained and compared.

Random Forest, XGBoost and CatBoost should be the primary directional models. The simpler models should be retained as baselines.

9 Eid and Holiday Direction Analysis

Directional forecasting is particularly useful around Eid. The model should use a signed Eid-relative-day feature:

$$\text{EidRelativeDay}_t = \text{Date}_t - \text{EidDate}. \quad (23)$$

For example:

Table 2: Models for directional forecasting

Model	Purpose
Persistence baseline	Uses the most recent directional value or share as the prediction.
Seasonal baseline	Uses the corresponding directional value from seven days earlier.
Linear Regression	Provides an interpretable linear baseline for directional prediction.
Random Forest	Captures nonlinear relationships among history, holidays, weather and direction.
XGBoost	Provides a strong boosted-tree model and can handle complex event-related patterns.
CatBoost	Provides an additional boosting comparison suitable for calendar, holiday and event variables.
Residual MLP	Predicts the change from the previous directional value or traffic share.

- -3 represents three days before Eid;
- 0 represents Eid day;
- $+3$ represents three days after Eid.

Separate directional profiles should be created for:

- Eid-ul-Fitr;
- Eid-ul-Adha;
- Days before Eid;
- Eid day;
- Days after Eid;
- Ordinary public holidays.

The analysis should determine:

- When the directional traffic surge begins;
- Which side records the larger traffic volume before Eid;
- Whether the dominant side changes after Eid;
- How quickly traffic returns to its normal directional distribution.

These relationships should be described as patterns associated with Eid travel unless the BBA side variables are confirmed to represent specific travel directions.

10 Training Procedure

The proposed update will follow these steps:

1. Load and clean the existing Padma Bridge dataset.
2. Verify the inconsistent Mawa, Jajira and total records against the original source reports.
3. Preserve the existing total traffic and toll forecasting models.
4. Create side-specific lag, rolling, share and directional difference features.
5. Generate targets for Mawa traffic, Jajira traffic, Mawa toll and Jajira toll.
6. Add future calendar, holiday, Eid and forecast-horizon features.
7. Divide the data chronologically into training, validation and test sets.
8. Train direct directional models.
9. Train total-and-share directional models.
10. Tune the models using expanding-window time-series cross-validation.
11. Compare direct and total-and-share approaches on the same test period.
12. Select the strongest directional model.
13. Save the models and update the prediction interface.

No random train-test split should be used because future observations must not influence past training data.

11 Expected Outputs

The updated system will generate the following output for every forecast date:

Table 3: Example directional forecast

Date	Mawa traffic	Jajira traffic	Mawa toll	Jajira toll
10 Sep 2026	11,800	9,400	BDT 13.7m	BDT 11.1m
11 Sep 2026	12,500	10,200	BDT 14.5m	BDT 12.0m

The system should also report:

- Predicted total traffic;
- Predicted total toll revenue;

- Mawa traffic percentage;
- Jajira traffic percentage;
- Dominant side;
- Whether the date is a weekend, holiday or Eid-related day;
- Weekly and monthly directional totals.

An example summary is:

Prediction date: 10 September 2026

Mawa-side traffic: 11,800 vehicles
 Jajira-side traffic: 9,400 vehicles
 Total traffic: 21,200 vehicles

Mawa traffic share: 55.7 percent
 Jajira traffic share: 44.3 percent

Mawa-side toll: BDT 13.7 million
 Jajira-side toll: BDT 11.1 million
 Total toll: BDT 24.8 million

12 Model Evaluation

The directional models will be evaluated using:

- Mean Absolute Error (MAE);
- Root Mean Squared Error (RMSE);
- Mean Absolute Percentage Error (MAPE);
- Coefficient of Determination (R^2);
- Directional-share MAE;
- Total-reconciliation error.

The reconciliation error for traffic may be defined as:

$$E_t^{\text{reconciliation}} = \left| \hat{Y}_t^{\text{total traffic}} - \left(\hat{Y}_t^{M,T} + \hat{Y}_t^{J,T} \right) \right|. \quad (24)$$

For the total-and-share method, this error should be zero.

Performance should be evaluated separately for:

Table 4: Structure of directional model comparison

Model	Mawa MAE	Jajira MAE	Mawa MAPE	Jajira MAPE	Share MAE
Linear Regression	—	—	—	—	—
Random Forest	—	—	—	—	—
XGBoost	—	—	—	—	—
CatBoost	—	—	—	—	—
Residual MLP	—	—	—	—	—

- Normal days;
- Weekends;
- Public holidays;
- Seven days before and after Eid;
- One-day forecasts;
- Seven-day forecasts;
- Thirty-day forecasts.

13 Comparison of Directional Strategies

The direct and total-and-share approaches should be compared using the following criteria:

Table 5: Comparison of proposed directional strategies

Criterion	Direct directional models	Total-and-share models
Side prediction	Predicted directly	Calculated from total and predicted share
Total consistency	Not automatically guaranteed	Automatically guaranteed
Model complexity	Four target models	Existing total models plus share models
Interpretation	Simple individual predictions	Shows both total demand and directional distribution
Recommended role	Comparison baseline	Primary proposed approach

14 Expected Limitations

The directional forecasting system will have the following limitations:

- Only approximately four years of daily traffic data are available.
- The number of Eid and major-holiday examples is limited.
- Some records contain inconsistencies between side values and published totals.
- Mawa and Jajira may identify toll-plaza sides rather than formally documented north-bound and southbound directions.
- Future weather forecasts may be unavailable for the complete thirty-day period.
- Unexpected events such as accidents, strikes, road closures and toll-rate changes are not represented.
- Thirty-day directional forecasts will generally be less accurate than next-day forecasts.

15 Expected Contribution

The proposed update will transform the existing total forecasting system into a more detailed operational system. Instead of reporting only the total expected bridge demand, it will show how that demand is distributed between the two sides.

This will be particularly valuable for:

- Eid traffic management;
- Toll-plaza staff allocation;
- Lane and traffic-control planning;
- Direction-specific demand monitoring;
- Weekly and monthly operational planning.

16 Conclusion

The directional forecasting update will separately predict Mawa- and Jajira-side traffic and toll revenue. New directional models must be trained because the existing models predict only bridge-wide totals.

Two approaches will be compared: direct directional forecasting and total-and-share forecasting. The total-and-share approach is recommended because it preserves consistency between the side predictions and the existing total forecasts.

Random Forest, XGBoost and CatBoost will be the primary models, while Linear Regression, persistence, seasonal baselines and the Residual MLP will provide additional

comparisons. The final system will support directional predictions for the next day, next seven days and next thirty days.

The main contribution can be summarized as follows:

This project extends total Padma Bridge forecasting into an Eid- and holiday-aware directional forecasting system that predicts Mawa- and Jajira-side traffic and toll revenue for daily, weekly and monthly planning.